

(12) **United States Patent**
Suzuki

(10) **Patent No.:** **US 12,404,935 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SLIDING COMPONENT**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 121 days.

(21) Appl. No.: **18/013,520**

(22) PCT Filed: **Jul. 1, 2021**

(86) PCT No.: **PCT/JP2021/024944**

§ 371 (c)(1),

(2) Date: **Dec. 28, 2022**

(87) PCT Pub. No.: **WO2022/009770**

PCT Pub. Date: **Jan. 13, 2022**

(65) **Prior Publication Data**

US 2023/0296131 A1 Sep. 21, 2023

(30) **Foreign Application Priority Data**

Jul. 6, 2020 (JP) 2020-116359

(51) **Int. Cl.**
F16J 15/34 (2006.01)

(52) **U.S. Cl.**
CPC **F16J 15/3416** (2013.01)

(58) **Field of Classification Search**
CPC .. F16J 15/3416; F16C 33/1065; F16C 17/045
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,876,515 A *	9/1932	Emmet		F16J 15/3416
				277/408
2,244,450 A *	6/1941	Erni		F16J 15/3416
				415/113
3,380,040 A	4/1968	Liggett		F16C 17/045
3,383,116 A	5/1968	Carter		277/96
3,443,813 A *	5/1969	Valdemar		F16J 15/3416
				277/408

(Continued)

FOREIGN PATENT DOCUMENTS

CN	1245552	2/2000		F16J 15/34
CN	2460801	11/2001		F16J 15/40

(Continued)

OTHER PUBLICATIONS

International Search Report and Written Opinion issued in PCT/JP2021/024940, dated Aug. 10, 2021, with English translation, 13 pages.

(Continued)

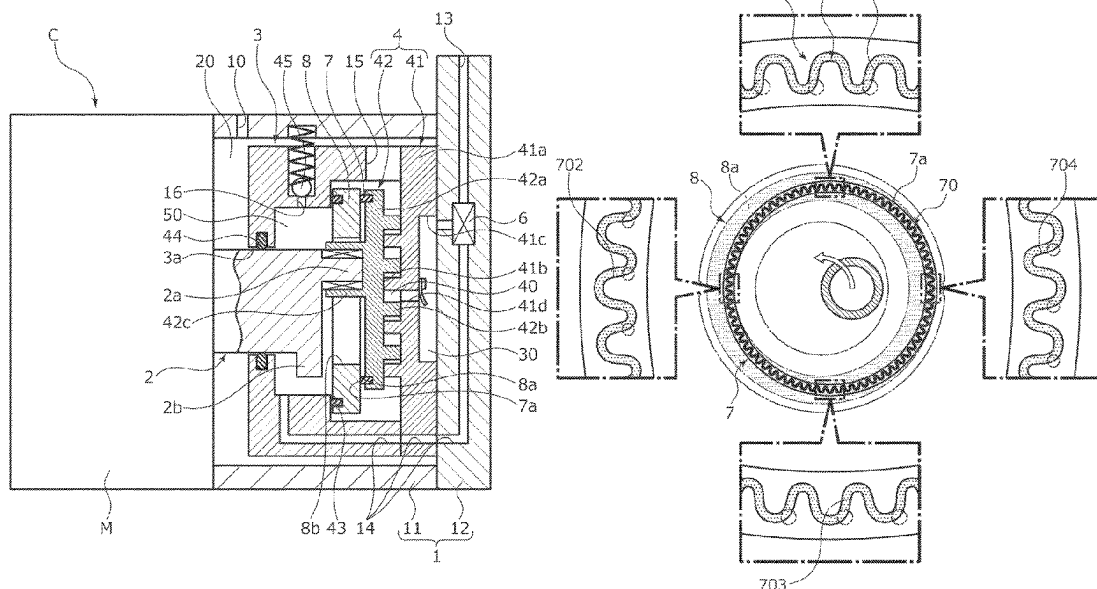
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(57) **ABSTRACT**

Provided is a sliding component capable of stably reducing the frictional resistance between sliding surfaces entailing eccentric rotation. A sliding component has a sliding surface relatively sliding with eccentric rotation. The sliding surface is provided with a dynamic pressure generation groove defined by side walls extending in a circumferential direction. At least one of the side walls is formed in a waveshape with amplitude in a radial direction.

11 Claims, 7 Drawing Sheets



References Cited

2007/0228664	A1	10/2007	Anand	F16J 15/3496
2007/0267820	A1	11/2007	Martin	F16J 15/3496
2007/0275267	A1	11/2007	Sabouni	F16J 15/3496
2007/0296156	A1	12/2007	Yanagisawa et al.	277/352
2008/0050260	A1	2/2008	Iwanami et al.	418/55.6
2008/0100001	A1	5/2008	Flaherty	
2009/0200749	A1	8/2009	Teshima	F16J 15/3484
2011/0194966	A1	8/2011	Takeuchi	F04B 18/0215
2011/0215531	A1	9/2011	Tokunaga et al.	277/399
2011/0215535	A1	9/2011	Vasagar	277/559
2011/0305871	A1	12/2011	Tabuchi	F16J 15/3284
2012/0018957	A1	1/2012	Watanabe	F16J 15/3/
2012/0217705	A1	8/2012	Hosoe	277/400
2013/0168928	A1	7/2013	Schrufer	C10M 103/02
2013/0209011	A1	8/2013	Tokunaga	
2013/0323105	A1	12/2013	Chao et al.	F04C 29/028
2014/0159314	A1*	6/2014	Hosoe	F16J 15/3496
				277/358
2014/0197600	A1	7/2014	Hosoe	F16J 15/342
2014/0217676	A1	8/2014	Hosoe et al.	277/350
2014/0294330	A1	10/2014	Tokunaga	F16C 17/04
2014/0319776	A1	10/2014	Theike et al.	
2015/0036955	A1	2/2015	Nomura et al.	F16C 33/20
2015/0115540	A1	4/2015	Tokunaga	F16J 15/34
2015/0123350	A1	5/2015	Itadani	F16J 15/3412
2015/0167847	A1	6/2015	Tokunaga	
2015/0184752	A1	7/2015	Itadani	F16J 15/3412
2015/0345642	A1	12/2015	Haas	F16J 15/3496
2015/0377297	A1	12/2015	Tokunaga et al.	F16C 33/748
2015/0377360	A1*	12/2015	Itadani	F16J 15/3448
				277/400
2016/0033045	A1	2/2016	Itadani et al.	F16J 15/3412
2016/0097457	A1	4/2016	Sun et al.	F16J 15/3412
2017/0089467	A1	3/2017	Young	F01D 11/003
2017/0146014	A1	5/2017	Ohta et al.	F04C 29/0021
2017/0241549	A1*	8/2017	Itadani	F16J 15/40
2017/0261107	A1	9/2017	Martin	F16J 15/3452
2017/0350407	A1	12/2017	Yamamoto et al.	F04D 29/046
2018/0017163	A1	1/2018	Hosoe et al.	F16J 15/164
2018/0073394	A1	3/2018	Tokunaga et al.	F01D 25/18
2018/0112711	A1	4/2018	Itadani	F16J 15/363
2018/0128377	A1	5/2018	Tukunaga et al.	F16J 15/342
2018/0128378	A1	5/2018	Tokunaga et al.	F16J 15/342
2018/0135699	A1	5/2018	Tokunaga et al.	F16C 33/80
2018/0172162	A1	6/2018	Tokunaga et al.	F16J 15/34
2018/0195618	A1	7/2018	Itadani et al.	F16J 15/3416
2018/0299015	A1	10/2018	Itadani	F16J 15/3448
2019/0169988	A1	6/2019	Tokunaga et al.	F01C 19/12
2019/0170257	A1	6/2019	Hosoe et al.	F16J 15/3412
2019/0285115	A1	9/2019	Negishi et al.	F16C 17/045
2019/0301522	A1	10/2019	Negishi et al.	F16C 17/02
2019/0331162	A1	10/2019	Negishi	F16J 15/34
2020/0141444	A1	5/2020	Thatte	F16J 17/08
2020/0224722	A1*	7/2020	Imura	F16C 33/743
2020/0224768	A1	7/2020	Imura	F16J 15/3412
2020/0240470	A1	7/2020	Sorgenti	F16J 15/3212
2020/0332901	A1	10/2020	Imura	F16J 15/3412
2021/0041026	A1	2/2021	Imura	F16J 15/3424
2021/0048062	A1	2/2021	Masumi et al.	F16C 17/102
2021/0048106	A1	2/2021	Imura et al.	F16J 15/3412
2021/0080009	A1	3/2021	Kimura et al.	F16J 15/3412
2021/0116030	A1	4/2021	Kimura et al.	F16J 15/182
2021/0116032	A1*	4/2021	Kimura	F16J 15/188
2021/0355992	A1*	11/2021	Tokunaga	F16J 15/3412
2021/0364034	A1	11/2021	Okada	F16J 15/342
2022/0056949	A1	2/2022	Ikeda et al.	F16C 17/04
2022/0128151	A1	4/2022	Tuk et al.	F16J

CN	1401924	3/2003		F16J	15/16
CN	1529063	9/2004		F16C	33/10
CN	1607710	4/2005		F16C	17/08
CN	101644333	2/2010		F16J	15/34
CN	201496542	6/2010		F16J	15/16
CN	101793169	8/2010		F01D	11/08
CN	101793324	8/2010		F16J	15/16

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN	101861485	10/2010	F16J 15/34	JP	2004360903	A *	12/2004	F04D 29/106
CN	203098871	7/2013	F16J 15/34	JP	2005-155894		6/2005	F16J 17/04
CN	103557229	2/2014	F16C 17/04	JP	2005-180652		7/2005	F16J 15/34
CN	103557334	2/2014	F16J 15/34	JP	2005-315391		11/2005	
CN	203641506	6/2014	F16J 15/16	JP	2005-337503		12/2005	F16J 15/34
CN	104169622	11/2014	F16J 15/34	JP	2006-9614		1/2006	F04C 18/08
CN	104321568	1/2015	F16C 33/72	JP	2006-77899		3/2006	F16J 15/34
CN	104685273	6/2015	F16J 15/34	JP	2006-90524		4/2006	F16C 17/02
CN	205244387	5/2016	F16J 15/16	JP	2006-183702		7/2006	F16C 17/04
CN	106029294	10/2016	B23K 26/364	JP	2006-316677		11/2006	F04C 18/02
CN	205877184	1/2017	F16J 15/16	JP	2007-162045		6/2007	
CN	205877198	1/2017	F16J 15/34	JP	2008-51018		3/2008	F04C 18/02
CN	106439023	2/2017	F16J 15/16	JP	2008-51030		3/2008	F04C 18/02
CN	107489770	12/2017	F16J 15/34	JP	2008-106940		5/2008	F16J 15/34
CN	108131386	6/2018	F16C 17/10	JP	2011-74931		4/2011	F16J 15/34
CN	109237042	1/2019	F16J 15/34	JP	2011-185292		9/2011	F16J 15/34
CN	110770456	2/2020	F16C 33/12	JP	2012-2295		1/2012	F16J 15/34
CN	110925426	3/2020	F16J 15/16	JP	2012-062534		3/2012	
CN	111656065	9/2020	F16J 15/34	JP	2012-82794		4/2012	F04C 18/02
DE	3223703	6/1982	F16J 15/34	JP	2012-122135		6/2012	C25D 15/02
DE	10048256	9/2000	F16C 33/06	JP	2013-167216		8/2013	F04C 18/02
DE	102008038396	2/2010		JP	2013-213545		10/2013	F16C 32/06
EP	0369295	11/1988	F16J 15/34	JP	2014-529052		10/2014	
EP	0589514	9/1992	B21J 5/12	JP	2015-063647		4/2015	
EP	0518681	12/1992	G11B 15/60	JP	2015-68330		4/2015	F04C 29/00
EP	0637706	8/1993	F16J 15/34	JP	5693599		4/2015	
EP	2138225	12/2009	B01J 13/20	JP	2015-183631		10/2015	F04C 2/10
EP	2754931	7/2014	F16J 15/34	JP	2016-61208		4/2016	F04C 18/02
EP	3112078	1/2017	B23K 26/364	JP	2016-80090		5/2016	F16J 15/34
EP	3196516	7/2017	F16J 15/34	JP	5960145		7/2016	F16J 15/34
EP	3217049	9/2017	F16J 15/34	JP	WO2019049847		3/2019	F16J 15/34
EP	3396186	10/2018	F16C 33/10	JP	WO2020110922		6/2020	F16J 15/34
EP	3575621	12/2019	F16C 33/10	NL	2022585	B1 *	8/2020	F16J 15/164
EP	3575643	12/2019	F16J 15/34	WO	WO2006051702		5/2006	F16J 15/34
EP	3650722	5/2020	F16C 33/12	WO	WO2011115073		9/2011	F16J 15/34
FR	2342440	9/1997	F16J 15/34	WO	WO2012046749		4/2012	F16J 15/34
GB	1509482	5/1978	F16C 33/10	WO	WO2013035503		3/2013	F16J 15/34
GB	2263952	8/1993	F16J 15/34	WO	WO2013053411		4/2013	
JP	S51-034974	3/1976		WO	WO2014061544		4/2014	
JP	S52-143571	10/1977	F16J 15/26	WO	WO2014148316		9/2014	F16J 15/34
JP	S7163770	10/1982	F16J 15/3412	WO	WO2014148317		9/2014	F16J 15/34
JP	S59-195253	12/1984	F16J 15/34	WO	WO2014174725		10/2014	F16J 15/34
JP	S59-195254	12/1984	F16J 15/34	WO	WO2015111707		7/2015	F16J 15/18
JP	S61-8402	1/1986	F01C 1/01	WO	WO2016035860		3/2016	F16J 15/34
JP	S63-134883	6/1988	F04C 18/02	WO	WO2016167262		10/2016	F16J 15/34
JP	S63-190975	8/1988	F16J 15/34	WO	WO2016186015		11/2016	F16J 15/34
JP	H02-16381	1/1990	F04C 18/02	WO	WO2016186019		11/2016	F16J 15/34
JP	H02-236067	9/1990	F16J 15/34	WO	WO2016186020		11/2016	F16J 15/34
JP	H02-136863	11/1990	F16J 15/34	WO	WO2016203878		12/2016	F16J 15/34
JP	H04-50559	2/1992	F16J 15/34	WO	WO2017002774		1/2017	F16J 15/34
JP	H04-337165	11/1992	F16J 15/34	WO	WO2017061406		4/2017	F16J 15/34
JP	H04-362289	12/1992	F04C 18/02	WO	WO2018025629		2/2018	F02B 55/02
JP	H05-60247	3/1993	F16J 15/34	WO	WO2018092742		5/2018	F16J 15/34
JP	H05-296248	11/1993		WO	WO2018105505		6/2018	F16J 33/12
JP	H05-90049	12/1993	F16J 15/34	WO	WO2018139231		8/2018	F16J 15/34
JP	H06-17941	1/1994	F16J 15/34	WO	WO2020129846		6/2020	F16C 17/04
JP	H06-117547	4/1994	F16J 15/34	WO	WO2020130087		6/2020	F16J 15/34
JP	H06-174107	6/1994	F16J 15/34	WO	WO2021125201		6/2021	F16J 15/18
JP	H6200927	7/1994	F16C 17/02					
JP	H06-323442	11/1994	F16J 15/34					
JP	H06-105105	12/1994	F16J 15/34					
JP	H07-43038	5/1995	F16J 15/34					
JP	9-89119	3/1997	F16J 15/34					
JP	H09228968	9/1997	F01C 17/06					
JP	9-292034	11/1997	F16J 15/34					
JP	H10-281299	10/1998	F16J 15/34					
JP	H10-292867	11/1998						
JP	H10-339286	12/1998	F04C 18/02					
JP	H11-132163	5/1999	F04C 18/02					
JP	H11-287329	10/1999	F16J 15/34					
JP	H11-303858	11/1999	F16C 17/10					
JP	3066367	5/2000	F16J 15/34					
JP	2001-12458	1/2001	F16C 17/10					
JP	2003-343730	12/2003	F16J 15/22					
JP	2004-360903	12/2004	F16J 15/447					

OTHER PUBLICATIONS

International Preliminary Report on Patentability issued in PCT/JP2021/024940, dated Jan. 10, 2023, 4 pages.

International Search Report and Written Opinion issued in PCT/JP2021/024941, dated Sep. 14, 2021, with English translation, 13 pages.

International Preliminary Report on Patentability issued in PCT/JP2021/024941, dated Jan. 10, 2023, 5 pages.

International Search Report and Written Opinion issued in PCT/JP2021/024938, dated Aug. 3, 2021, with English translation, 12 pages.

International Preliminary Report on Patentability issued in PCT/JP2021/024938, dated Jan. 10, 2023, 4 pages.

International Search Report and Written Opinion issued in PCT/JP2021/024945, dated Sep. 7, 2021, with English translation, 16 pages.

International Preliminary Report on Patentability issued in PCT/JP2021/024945, dated Jan. 10, 2023, 6 pages.

(56)

References Cited

OTHER PUBLICATIONS

International Search Report and Written Opinion issued in PCT/JP2021/024943, dated Sep. 7, 2021, with English translation, 13 pages.
 International Preliminary Report on Patentability issued in PCT/JP2021/024943, dated Jan. 10, 2023, 5 pages.
 International Search Report and Written Opinion issued in PCT/JP2021/024944, dated Jul. 27, 2021, with English translation, 11 pages.
 International Preliminary Report on Patentability issued in PCT/JP2021/024944, dated Jan. 10, 2023, 4 pages.
 Official Action issued in related U.S. Appl. No. 18/012,853, dated Sep. 8, 2023, 11 pages.
 Official Action issued in related U.S. Appl. No. 18/012,857, dated Oct. 5, 2023, 11 pages.
 Official Action issued in related U.S. Appl. No. 18/012,856, dated Sep. 11, 2023, 8 pages.
 Notice of Allowance issued in related U.S. Appl. No. 18/013,515, dated Dec. 19, 2023, 10 pages.
 Official Action issued in related U.S. Appl. No. 18/012,856, dated Feb. 2, 2024, 9 pages.

Official Action issued in related U.S. Appl. No. 18/013,513, dated Feb. 15, 2024, 9 pages.
 Official Action issued in related U.S. Appl. No. 18/012,857, dated Mar. 21, 2024, 13 pages.
 Japanese Official Action issued in related application serial No. 2022-535275, dated Feb. 13, 2024, 8 pages.
 Chinese Official Action issued in related Chinese Patent Application Serial No. 202180044553.5, dated Apr. 28, 2024, 13 pages with translation.
 European Search Report issued in European Patent Application Serial No. 21837223.3, dated Jun. 25, 2024, 7 pages.
 European Search Report issued in European Patent Application Serial No. 21837221.7, dated Jul. 2, 2024, 7 pages.
 European Search Report issued in European Patent Application Serial No. 21836928.8, dated Jun. 25, 2024, 8 pages.
 European Search Report issued in European Patent Application Serial No. 21837222.5, dated Jul. 12, 2024, 7 pages.
 Korean Official Action issued in related Korean Patent Application Serial No. 10-2023-7000902, dated Jul. 26, 2024, 9 pages with translation.
 Korean Official Action issued in related Korean Patent Application Serial No. 10-2023-7000900, dated Jul. 26, 2024, 8 pages with translation.

* cited by examiner

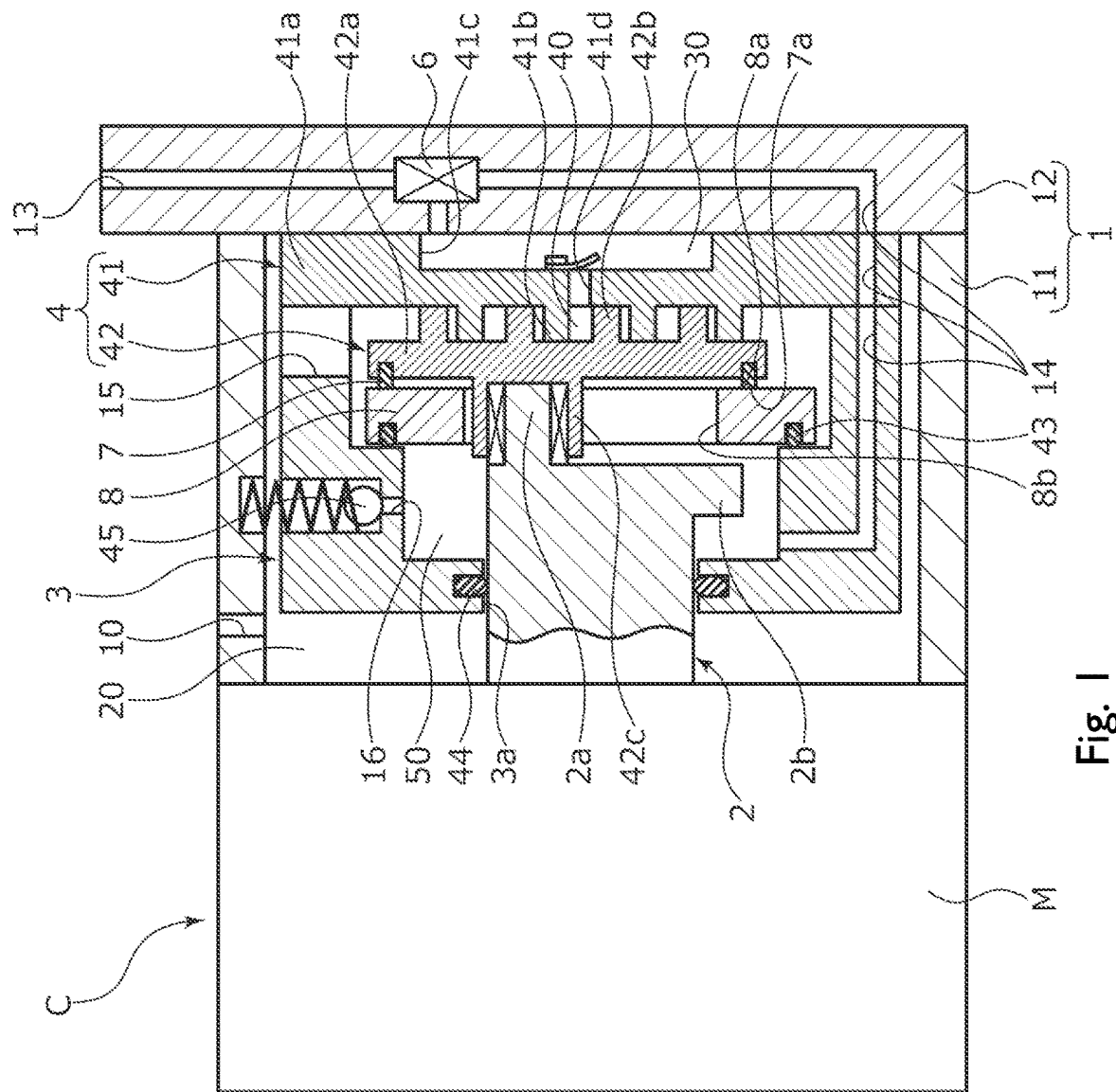


Fig.2

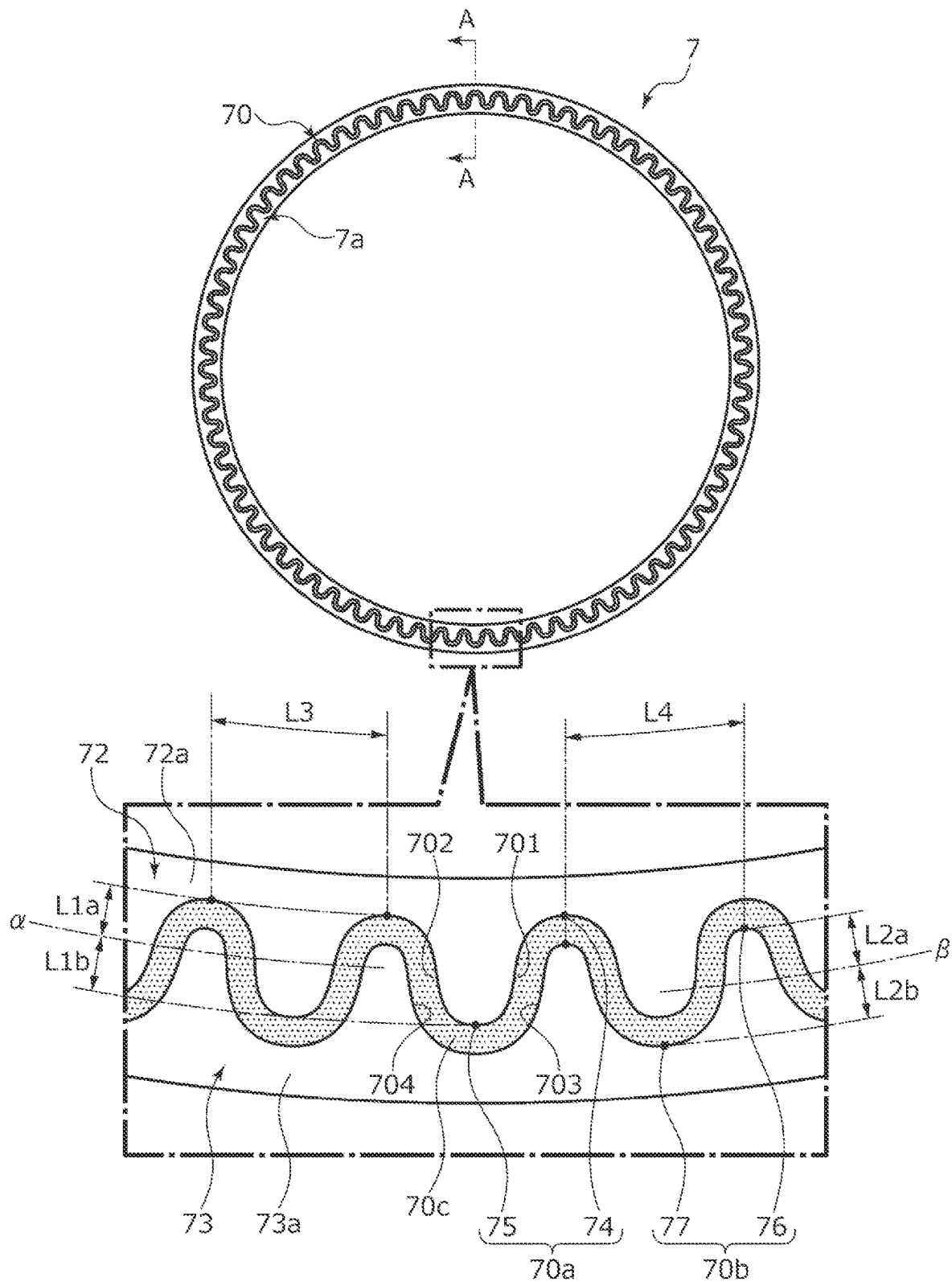
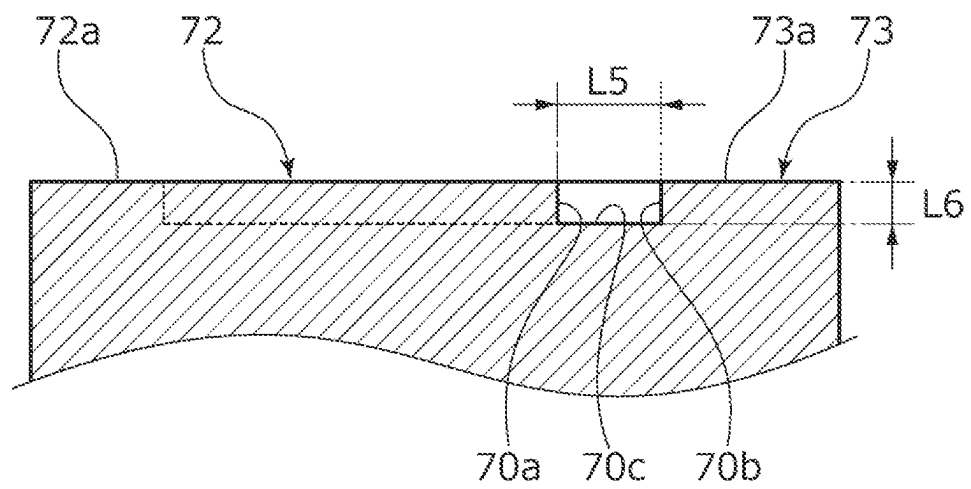


Fig.3



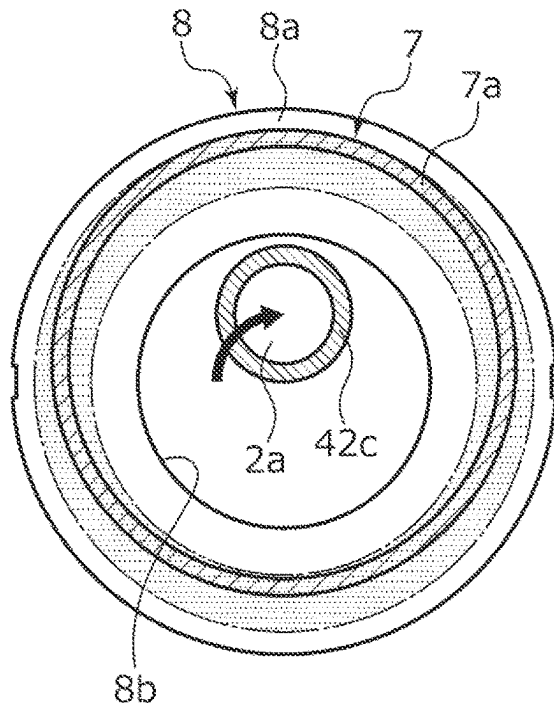


Fig. 4(a)

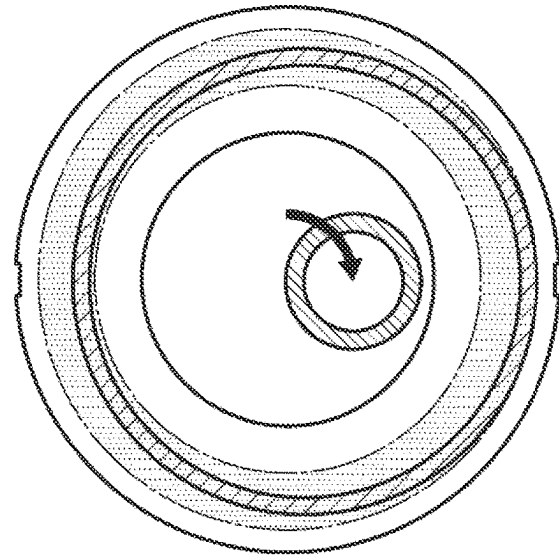


Fig. 4(b)

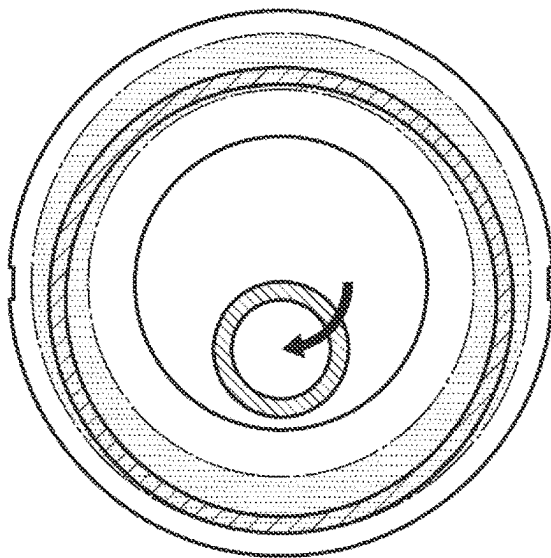


Fig. 4(c)

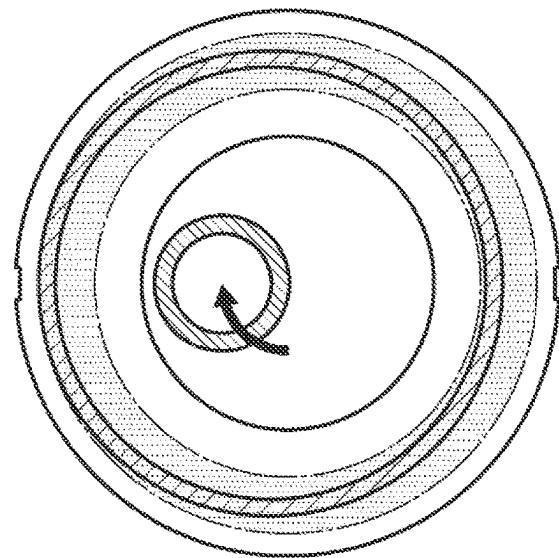


Fig. 4(d)

Fig.5

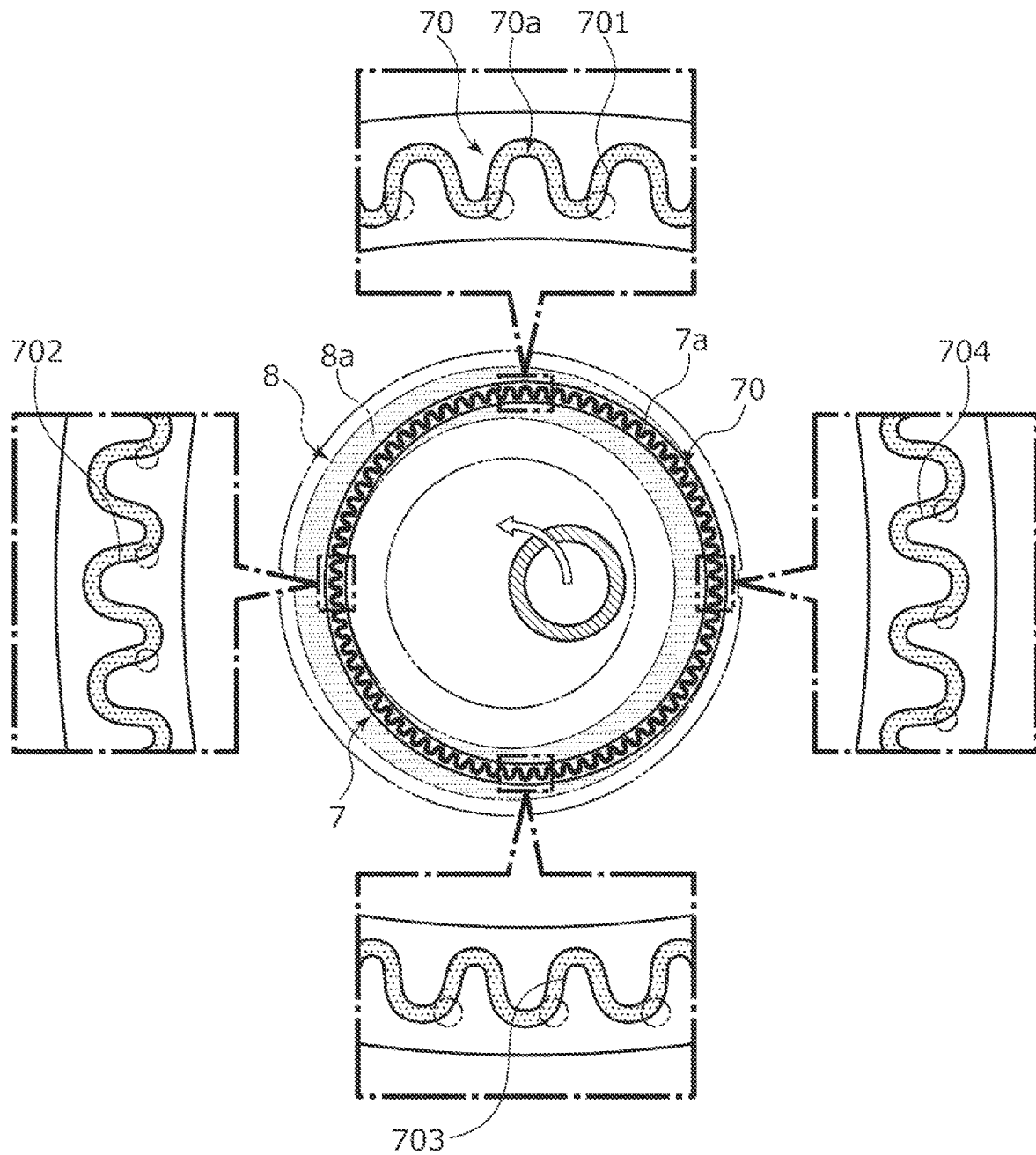


Fig.6

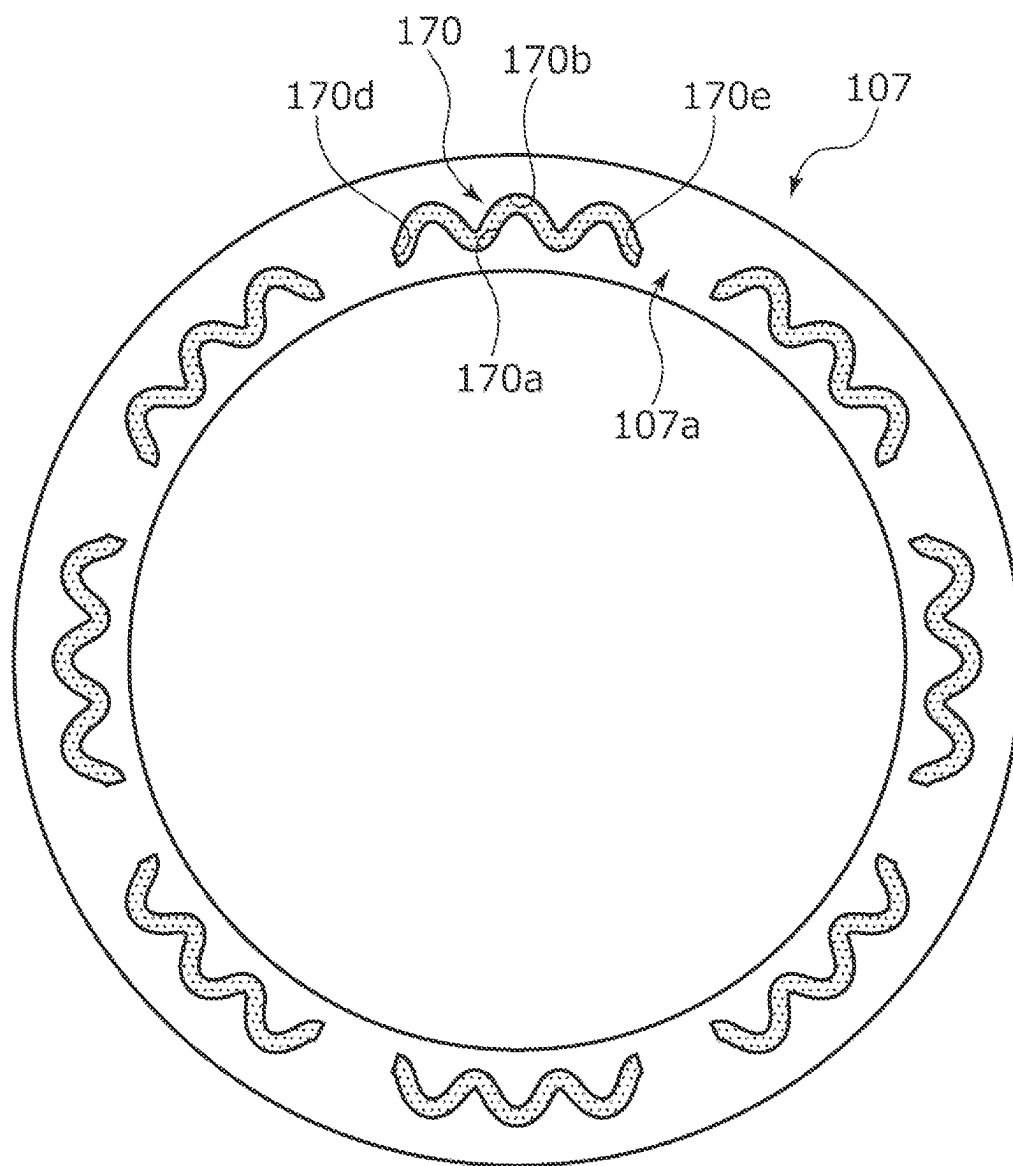
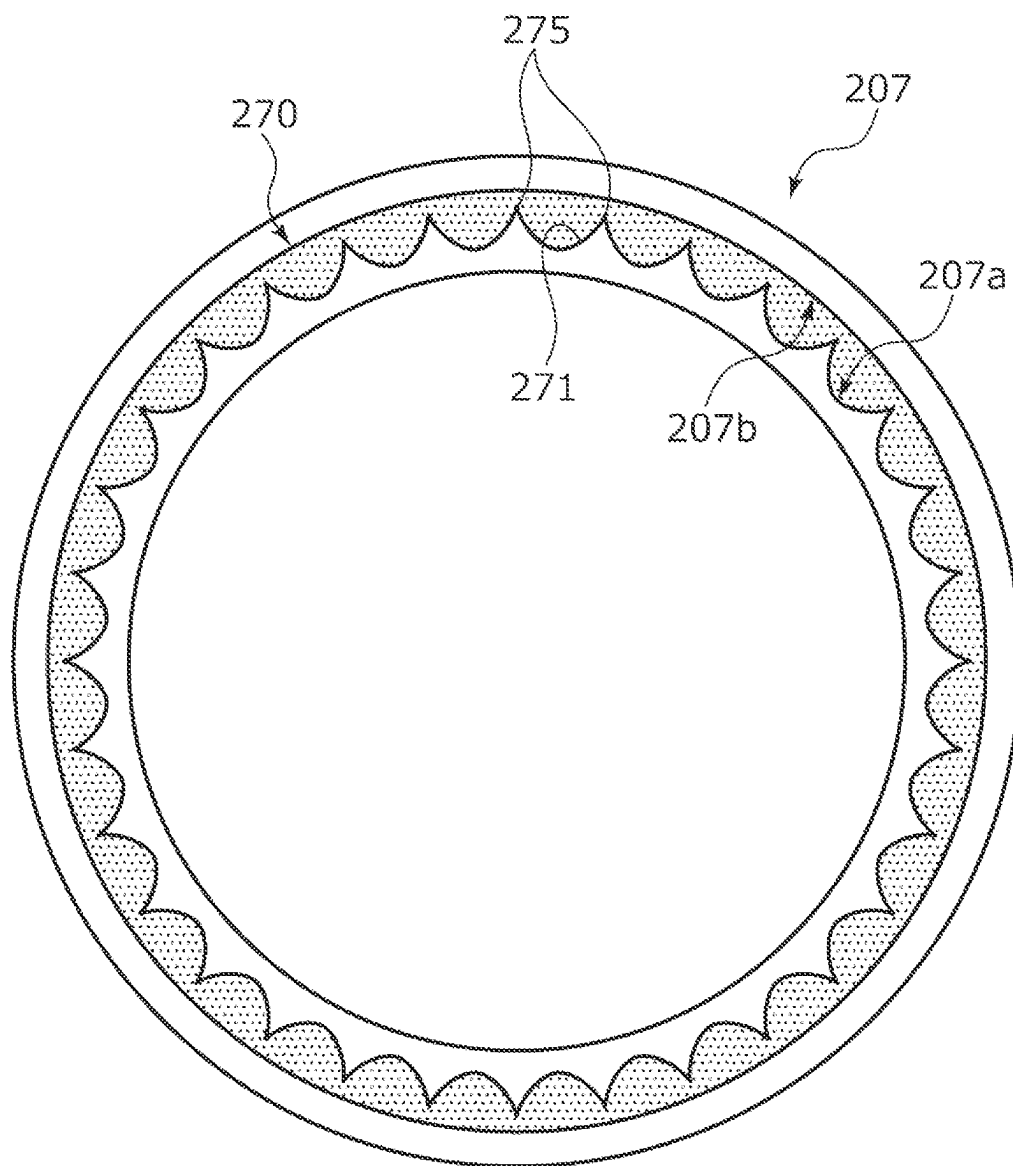


Fig.7



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SLIDING COMPONENT

TECHNICAL FIELD

The present invention relates to a sliding component used in a rotating machine including an eccentric mechanism.

BACKGROUND ART

Machines entailing rotational driving and used in various industrial fields include not only a rotating machine rotating with its central axis held at a fixed position but also a rotating machine rotating with its central axis entailing eccentricity. The rotating machine rotating with eccentricity is, for example, a scroll compressor. This type of compressor is provided with, for example, a scroll compression mechanism including a fixed scroll having a spiral lap on the surface of an end plate and a movable scroll having a spiral lap on the surface of an end plate and an eccentric mechanism eccentrically rotating a rotary shaft. By the rotary shaft rotating, the movable scroll is slid relative to the fixed scroll with eccentric rotation. As a result, in this mechanism, the fluid supplied from the low-pressure chamber on the outer diameter side of the two scrolls is pressurized and a high-pressure fluid is discharged from the discharge hole formed in the middle of the fixed scroll.

These scroll compressors using the mechanism in which the movable scroll is slid relative to the fixed scroll with eccentric rotation are widely used in, for example, refrigeration cycles because the compressors are highly efficient in terms of compression and cause little noise. However, the compressors are problematic in that a refrigerant leaks from the axial gap between the two scrolls. The scroll compressor described in Patent Citation 1 includes a thrust plate sliding relative to a movable scroll on the back surface side of the movable scroll. A part of a refrigerant compressed by a scroll compression mechanism is supplied to the back pressure chamber formed on the back surface side of the thrust plate, and the movable scroll is pressed toward a fixed scroll. As a result, it is possible to reduce the leakage of the refrigerant from the axial gap between the two scrolls when the refrigerant is compressed.

CITATION LIST

Patent Literature

Patent Citation 1: JP 2016-61208 A (Pages 5 to 6, FIG. 1)

SUMMARY OF INVENTION

Technical Problem

In the scroll compressor described in Patent Citation 1, a part of the refrigerant compressed by the scroll compression mechanism is used to press the movable scroll from the back surface side toward the fixed scroll via the thrust plate. Accordingly, although the refrigerant leakage from the inter-scroll axial gap can be reduced, a pressing force acts from both axial sides between the two scrolls, especially on the sliding surface that entails the eccentric rotation between the movable scroll and the thrust plate. As a result, there is a problem that an increase in frictional resistance occurs, a smooth operation of the movable scroll is hindered, and the efficiency of compression cannot be enhanced.

The present invention has been made in view of such a problem, and an object of the present invention is to provide

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a sliding component capable of stably reducing the frictional resistance between sliding surfaces entailing eccentric rotation.

Solution to Problem

In order to solve the above problem, a sliding component according to the present invention is a sliding component having a sliding surface relatively sliding with eccentric rotation, wherein the sliding surface is provided with a dynamic pressure generation groove defined by side walls extending in a circumferential direction, at least one of the side walls being formed in a waveshape with amplitude in a radial direction. According to the aforesaid feature of the present invention, the side wall of the dynamic pressure generation groove extends in the circumferential direction in a waveshape having an amplitude in the radial direction. Accordingly, it is possible to ensure a plurality of intersection surfaces generating dynamic pressure regardless of the eccentric rotation angle. As a result, the sliding surface can be separated from another sliding surface in a state where the inclination thereof is reduced, and the frictional resistance between the sliding surfaces can be stably reduced.

It may be preferable that both the side walls of the dynamic pressure generation groove are formed in the waveshape with the amplitude in the radial direction. According to this preferable configuration, when the sliding surface relatively slides with eccentric rotation, dynamic pressure can be generated at a plurality of points in the circumferential direction on both the side wall on the inner diameter side and the side wall on the outer diameter side. Accordingly, the sliding surface is separated in a state where the inclination thereof is reduced.

It may be preferable that the dynamic pressure generation groove is formed in an annular shape. According to this preferable configuration, dynamic pressure can be generated over the entire circumference of the sliding surface.

It may be preferable that the waveshape of the side wall is periodically formed with a constant pitch. According to this preferable configuration, dynamic pressure can be equally generated over the circumferential direction of the sliding surface.

It may be preferable that the waveshape of the side wall is formed by curved surfaces. According to this preferable configuration, the movement of the fluid that moves in the dynamic pressure generation groove becomes smooth and dynamic pressure can be stably generated.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic configuration diagram illustrating a scroll compressor to which a side seal as a sliding component according to a first embodiment of the present invention is applied.

FIG. 2 is a diagram illustrating a sliding surface of the side seal in the first embodiment of the present invention.

FIG. 3 is an A-A cross-sectional view.

FIG. 4 is a diagram illustrating relative sliding between the sliding surface of the side seal and a sliding surface of a thrust plate in the first embodiment of the present invention. It should be noted that FIG. 4A illustrates the start position of the relative sliding and FIGS. 4B to 4D illustrate the positional relationships between the relatively sliding surfaces of the side seal and the thrust plate at an eccentric rotary shaft rotation of 90, 180, and 270 degrees, respectively.

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FIG. 5 is a diagram illustrating where the dynamic pressure generated in a dynamic pressure generation groove is generated on the sliding surface of the side seal eccentrically rotating from the state of FIG. 4A toward the state of FIG. 4B.

FIG. 6 is a diagram illustrating a sliding surface of a side seal as a sliding component according to a second embodiment of the present invention.

FIG. 7 is a diagram illustrating a sliding surface of a side seal as a sliding component according to a third embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

Modes for implementing the sliding component according to the present invention will be described below based on embodiments.

First Embodiment

The sliding component according to a first embodiment of the present invention will be described with reference to FIGS. 1 to 5. It should be noted that in the drawings, the groove formed in the sliding surface of the sliding component or the like is dotted for convenience of description.

The sliding component according to the first embodiment of the present invention is applied to a rotating machine including an eccentric mechanism such as a scroll compressor C that suctions, compresses, and discharges a refrigerant as a fluid used in the air conditioning system of an automobile or the like. It should be noted that the refrigerant in the present embodiment is a gas mixed with a mist-like lubricating oil.

First, the scroll compressor C will be described. As illustrated in FIG. 1, the scroll compressor C mainly includes a housing 1, a rotary shaft 2, an inner casing 3, a scroll compression mechanism 4, a side seal 7 as a sliding component, a thrust plate 8, and a drive motor M.

The housing 1 includes a cylindrical casing 11 and a cover 12 blocking an opening of the casing 11. The drive motor M blocks the opening in the casing 11 that is on the side axially opposite to the opening in the casing 11 that is blocked by the cover 12.

Formed in the casing 11 are a low-pressure chamber 20, a high-pressure chamber 30, and a back pressure chamber 50. A low-pressure refrigerant is supplied from a refrigerant circuit (not illustrated) to the low-pressure chamber 20 as the external space on the low-pressure side through a suction port 10. A high-pressure refrigerant compressed by the scroll compression mechanism 4 is discharged to the high-pressure chamber 30. A part of the refrigerant compressed by the scroll compression mechanism 4 is supplied, together with lubricating oil, to the back pressure chamber 50 as the external space on the high-pressure side. It should be noted that the back pressure chamber 50 is formed in the cylindrical inner casing 3 accommodated in the casing 11.

A discharge communication passage 13 is formed in the cover 12. The discharge communication passage 13 allows the refrigerant circuit (not illustrated) and the high-pressure chamber 30 to communicate with each other. In addition, a part of a back pressure communication passage 14 for communication between the high-pressure chamber 30 and the back pressure chamber 50 is formed in the cover 12 by branching off from the discharge communication passage 13. It should be noted that the discharge communication passage 13 is provided with an oil separator 6 for lubricating oil separation from a refrigerant.

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The inner casing 3 is fixed with an axial end portion of the inner casing 3 abutting against an end plate 41a of a fixed scroll 41 constituting the scroll compression mechanism 4. In addition, the inner casing 3 has a side wall where a suction communication passage 15 penetrating the wall in the radial direction is formed. In other words, the low-pressure chamber 20 is formed from the outside of the inner casing 3 to the inside of the inner casing 3 via the suction communication passage 15. The refrigerant supplied to the inside of the inner casing 3 through the suction communication passage 15 is suctioned into the scroll compression mechanism 4.

The scroll compression mechanism 4 mainly includes the fixed scroll 41 and a movable scroll 42. The fixed scroll 41 is fixed to the cover 12 in a sealed shape. The movable scroll 42 is accommodated in the inner casing 3.

The fixed scroll 41 is made of metal and includes a spiral lap 41b. The spiral lap 41b projects toward the movable scroll 42 from the surface of the disk-shaped end plate 41a, that is, the end plate 41a. In addition, the fixed scroll 41 has a recessed portion 41c where the inner diameter side of the back surface of the end plate 41a, that is, the end surface of the end plate 41a that abuts against the cover 12 is recessed in the direction opposite to the cover 12. The high-pressure chamber 30 is defined from the recessed portion 41c and the cover 12.

The movable scroll 42 is made of metal and includes a spiral lap 42b. The spiral lap 42b projects toward the fixed scroll 41 from the surface of a disk-shaped end plate 42a, that is, the end plate 42a. In addition, a boss 42c protruding from the middle of the back surface of the end plate 42a is formed on the movable scroll 42. An eccentric portion 2a formed on the rotary shaft 2 is fitted into the boss 42c so as to be relatively rotatable. It should be noted that an eccentric mechanism causing the rotary shaft 2 to perform eccentric rotation is configured by the eccentric portion 2a of the rotary shaft 2 and a counterweight portion 2b protruding in the outer diameter direction from the rotary shaft 2 in the present embodiment.

When the rotary shaft 2 is rotationally driven by the drive motor M, the eccentric portion 2a rotates eccentrically and the movable scroll 42 slides, in a posture-maintained state, relative to the fixed scroll 41 with the eccentric rotation. At this time, the movable scroll 42 rotates eccentrically with respect to the fixed scroll 41 and, with this rotation, the contact positions of the laps 41b and 42b sequentially move in the rotation direction and a compression chamber 40 formed between the laps 41b and 42b gradually shrinks while moving toward the middle. As a result, the refrigerant suctioned into the compression chamber 40 from the low-pressure chamber 20 formed on the outer diameter side of the scroll compression mechanism 4 is compressed and, finally, the high-pressure refrigerant is discharged to the high-pressure chamber 30 through a discharge hole 41d provided in the middle of the fixed scroll 41.

Next, the side seal 7 as a sliding component in the present embodiment will be described. As illustrated in FIGS. 2 and 3, the side seal 7 is made of resin and has a rectangular cross section and an annular shape in an axial view. In addition, the side seal 7 is fixed to the back surface of the end plate 42a of the movable scroll 42 (see FIG. 1). It should be noted that a sliding surface 7a of the side seal 7 is illustrated in FIG. 2.

The side seal 7 has the sliding surface 7a abutting against a sliding surface 8a (see FIG. 1) of the thrust plate 8.

As illustrated in FIG. 2, the sliding surface 7a of the side seal 7 includes a dynamic pressure generation groove 70 formed in an endless ring shape along the sliding surface 7a.

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As illustrated in FIGS. 2 and 3, the dynamic pressure generation groove 70 is partitioned by an inner side wall 70a, an outer side wall 70b, and a bottom surface 70c. The inner side wall 70a is formed as a side wall on the inner diameter side extending in the depth direction orthogonal to a flat surface 72a of a land 72 on the inner diameter side. The outer side wall 70b is formed as a side wall on the outer diameter side extending in the depth direction orthogonal to a flat surface 73a of a land 73 on the outer diameter side. The bottom surface 70c extends parallel to the surfaces 72a and 73a formed on the same surface and is formed by connecting the end portions of the inner side wall 70a and the outer side wall 70b.

The inner side wall 70a and the outer side wall 70b have a waveshape extending in the circumferential direction and having an amplitude in the radial direction when viewed from the axial direction. The inner side wall 70a and the outer side wall 70b have a waveshape with a constant period and a constant amplitude.

It should be noted that in the present embodiment below, the radial apexes in the amplitude of the inner side wall 70a are referred to as an inner diameter side apex 74 and an outer diameter side apex 75, and the radial apexes in the amplitude of the outer side wall 70b are referred to as an inner diameter side apex 76 and an outer diameter side apex 77.

Further, it should be noted that the part extending in an S shape from the inner diameter side apex 74 of the inner side wall 70a toward the outer diameter side apex 75 adjacent to the outer diameter side in the clockwise direction with respect to the middle of the side seal 7 is referred to as an intersection surface 701. In addition, the part extending in an inverted S shape from the inner diameter side apex 74 toward the outer diameter side apex 75 adjacent to the outer diameter side in the counterclockwise direction with respect to the middle of the side seal 7 is referred to as an intersection surface 702. In addition, the part extending in an S shape from the inner diameter side apex 76 of the outer side wall 70b toward the outer diameter side apex 77 adjacent to the outer diameter side in the clockwise direction with respect to the middle of the side seal 7 is referred to as an intersection surface 703. In addition, the part extending in an inverted S shape from the inner diameter side apex 76 toward the outer diameter side apex 77 adjacent to the outer diameter side in the counterclockwise direction with respect to the middle of the side seal 7 is referred to as an intersection surface 704.

Specifically, as illustrated in FIG. 2, the inner diameter side apexes 74, the outer diameter side apexes 75, the inner diameter side apexes 76, and the outer diameter side apexes 77 each are equally arranged in the circumferential direction. As for the inner diameter side apex 74 and the outer diameter side apex 75 adjacent to each other on the inner side wall 70a, a maximum amplitude L1a on the inner diameter side and a maximum amplitude L1b on the outer diameter side are equal with respect to a virtual reference line extending on the circumference, and these maximum amplitudes L1a and L1b are equal over the circumferential direction.

Further, as for the inner diameter side apex 76 and the outer diameter side apex 77 adjacent to each other on the outer side wall 70b, a maximum amplitude L2a on the inner diameter side and a maximum amplitude L2b on the outer diameter side are equal with respect to a virtual reference line β extending on the circumference, and these maximum amplitudes L2a and L2b are equal over the circumferential direction. It should be noted that the maximum amplitudes L1a and L1b and the maximum amplitudes L2a and L2b have the same length (i.e., $L1a, L1b=L2a, L2b$).

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In addition, a circumferential distance L3 (that is, wavelength) between the inner diameter side apexes 74 adjacent on the inner side wall 70a is constant over the circumferential direction, and a circumferential distance L4 (that is, wavelength) between the inner diameter side apexes 76 adjacent on the outer side wall 70b is constant over the circumferential direction.

In addition, the distance L3 is substantially equal to the distance L4. Strictly speaking, the distance L4 is slightly longer than the distance L3 in terms of length ($L3=L4$). It should be noted that although not illustrated, the circumferential distance between the outer diameter side apexes 75 is the same length as the distance L3 and the circumferential distance between the outer diameter side apexes 77 is the same length as the distance L4.

In addition, the inner side wall 70a and the outer side wall 70b have a waveshape having a curved surface shape. Specifically, the part near the inner diameter side apex 74 and the inner diameter side apex 76 is a curved surface convex to the inner diameter side, and the part near the outer diameter side apex 75 and the outer diameter side apex 77 is a curved surface convex to the outer diameter side.

As illustrated in FIG. 3, a radial width dimension L5 of the dynamic pressure generation groove 70 (that is, the separation width between the inner side wall 70a and the outer side wall 70b) is constant over the circumferential direction and formed larger than a depth dimension L6 of the dynamic pressure generation groove 70 (i.e., $L5>L6$). The radial width dimension L5 is constant over the entire circumference of the dynamic pressure generation groove 70. It should be noted that the width dimension L5 is preferably 10 times or more the depth dimension L6 although the width dimension and the depth dimension of the dynamic pressure generation groove 70 can be freely changed on condition that the width dimension of the dynamic pressure generation groove 70 is formed larger than the depth dimension of the dynamic pressure generation groove 70. It should be noted that for convenience of description, the depth dimension L6 of the dynamic pressure generation groove 70 illustrated in FIG. 3 is deeper than the actual depth dimension L6.

Referring to FIG. 1, the thrust plate 8 is made of metal and has an annular shape. In addition, a seal ring 43 is fixed to the thrust plate 8. The seal ring 43 abuts against the inside surface of the inner casing 3. As a result, the thrust plate 8 functions as a thrust bearing that receives an axial load of the movable scroll 42 via the side seal 7.

In addition, the side seal 7 and the seal ring 43 partition the low-pressure chamber 20 formed on the outer diameter side of the movable scroll 42 and the back pressure chamber 50 formed on the back surface side of the movable scroll 42 in the inner casing 3. The back pressure chamber 50 is a closed space formed between the inner casing 3 and the rotary shaft 2. A seal ring 44 is fixed to the inner periphery of a through hole 3a provided in the middle of the other end of the inner casing 3 and is in sliding contact in a sealed shape with the rotary shaft 2 inserted through the through hole 3a. In addition, the back pressure communication passage 14 allowing the high-pressure chamber 30 and the back pressure chamber 50 to communicate with each other is formed over the cover 12, the fixed scroll 41, and the inner casing 3. In addition, the back pressure communication passage 14 is provided with an orifice (not illustrated) and, after depressurization adjustment by means of the orifice, the refrigerant in the high-pressure chamber 30 is supplied to the back pressure chamber 50 together with the lubricating oil separated by the oil separator 6. At this time, the pressure in the back pressure chamber 50 is adjusted to be

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higher than the pressure in the low-pressure chamber 20. It should be noted that a pressure release hole 16 is formed in the inner casing 3, penetrates the inner casing 3 in the radial direction, and allows the low-pressure chamber 20 and the back pressure chamber 50 to communicate with each other. In addition, a pressure adjustment valve 45 is provided in the pressure release hole 16. The pressure adjustment valve 45 is opened by the pressure of the back pressure chamber 50 exceeding a set value.

In addition, the boss 42c of the movable scroll 42 is inserted through a through hole 8b in the middle of the thrust plate 8. The through hole 8b is formed to have a diameter size at which it is possible to allow eccentric rotation by the eccentric portion 2a of the rotary shaft 2 fitted into the boss 42c. In other words, the sliding surface 7a of the side seal 7 is capable of sliding relative to the sliding surface 8a of the thrust plate 8 with eccentric rotation by the eccentric rotation of the rotary shaft 2 (see FIG. 4).

It should be noted that FIGS. 4A to 4D in FIG. 4 illustrate the rotational trajectory of the boss 42c that is viewed from the fixed scroll 41 side and indicated by the black arrows. Respectively illustrated in FIGS. 4B to 4D are the boss 42c rotated by 90 degrees, 180 degrees, and 270 degrees with FIG. 4A serving as the clockwise reference. In addition, the sliding region between the sliding surface 7a of the side seal 7 and the sliding surface 8a of the thrust plate 8 is schematically illustrated by dots. In addition, regarding the rotary shaft 2, the counterweight portion 2b constituting the eccentric mechanism and so on are not illustrated and only the eccentric portion 2a fitted into the boss 42c is illustrated for convenience of description.

As described above, the side seal 7 is a sliding component having the sliding surface 7a sliding relative to the sliding surface 8a of the thrust plate 8 with the eccentric rotation.

Next, dynamic pressure generation during the sliding of the side seal 7 relative to the thrust plate 8 will be described with reference to FIG. 5. It should be noted that FIG. 5 illustrates an aspect when the side seal 7 moves from the state of FIG. 4D toward the state of FIG. 4A. In addition, in FIG. 5, the side seal 7 in a case where the sliding surface 7a is viewed from the axial direction is illustrated, and the circle marks illustrated in the enlarged portions indicate the points where the pressure increases in the dynamic pressure generation groove 70. In addition, a fluid containing, for example, a refrigerant and lubricating oil is stored in the dynamic pressure generation groove 70 even when the rotation is stopped.

As illustrated in FIG. 5, when the side seal 7 is to move in the direction of the white arrow, the fluid in the dynamic pressure generation groove 70 moves in the opposite direction relative to the white arrow. As a result, dynamic pressure is generated on the inner side wall 70a and the outer side wall 70b partitioning the dynamic pressure generation groove 70, the sliding surfaces 7a and 8a are slightly separated from each other, and a fluid film is formed by the fluid.

Specifically, when the thrust plate 8 is regarded as an analog clock, the position directly above the paper surface is set to the 12 o'clock position and, at the 12 o'clock position in the dynamic pressure generation groove 70, dynamic pressure is mainly generated on each intersection surface 701 on the inner side wall 70a. In addition, at the 3 o'clock position in the dynamic pressure generation groove 70, dynamic pressure is mainly generated on each intersection surface 704 on the outer side wall 70b. In addition, at the 6 o'clock position in the dynamic pressure generation groove 70, dynamic pressure is mainly generated on each intersec-

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tion surface 703 on the outer side wall 70b. In addition, at the 9 o'clock position in the dynamic pressure generation groove 70, dynamic pressure is mainly generated on each intersection surface 702 on the inner side wall 70a.

As described above, the inner side wall 70a and the outer side wall 70b of the dynamic pressure generation groove 70 extend in the circumferential direction in a waveshape and have the plurality of intersection surfaces 701 to 704 intersecting with respect to the eccentric rotation of the side seal 7. Accordingly, dynamic pressure can be generated by the intersection surfaces 701 to 704 regardless of the eccentric rotation angle of the side seal 7, the sliding surfaces 7a and 8a can be separated from each other with the inclination of the sliding surfaces 7a and 8a reduced, and the frictional resistance of the sliding surfaces 7a and 8a can be stably reduced.

In addition, at the 12 o'clock and 6 o'clock positions in the dynamic pressure generation groove 70, the intersection surfaces 701 and 703 have a concave shape holding the fluid in the dynamic pressure generation groove 70 that moves toward the opposite direction relative to the white arrow. Accordingly, the fluid near the intersection surfaces 701 and 703 is unlikely to be dispersed in the circumferential direction, and dynamic pressure is generated with ease. In other words, a larger dynamic pressure is generated at the 12 o'clock and 6 o'clock positions in the dynamic pressure generation groove 70 than at the 3 o'clock and 9 o'clock positions in the dynamic pressure generation groove 70.

It should be noted that FIG. 5 illustrates a form when the side seal 7 moves from the state of FIG. 4D toward the state of FIG. 4A, dynamic pressure is generated in substantially the same form when the side seal 7 moves from the state of FIG. 4D toward the state of FIG. 4A, when the side seal 7 moves from the state of FIG. 4A toward the state of FIG. 4B, when the side seal 7 moves from the state of FIG. 4B toward the state of FIG. 4C, and when the side seal 7 moves from the state of FIG. 4C toward the state of FIG. 4D, and thus description is omitted as to the forms when the side seal 7 moves from the state of FIG. 4A toward the state of FIG. 4B, from the state of FIG. 4B toward the state of FIG. 4C, and from the state of FIG. 4C toward the state of FIG. 4D.

In addition, in the sliding of the side seal 7 relative to the thrust plate 8 and entailing eccentric rotation, the position where a large dynamic pressure is generated in the dynamic pressure generation groove 70 continuously moves along the circumferential direction of the dynamic pressure generation groove 70 in accordance with the movement direction of the sliding surface 7a of the side seal 7. Accordingly, a state where the relative inclination between the sliding surfaces 7a and 8a is suppressed can be maintained.

In addition, the dynamic pressure generation groove 70 is formed in an annular shape, and thus dynamic pressure can be generated over the entire circumference of the sliding surface 7a and the inclination of the sliding surfaces 7a and 8a can be reduced regardless of the eccentric rotation angle of the side seal 7.

In addition, the inner side wall 70a and the outer side wall 70b have a waveshape with a constant period, and thus dynamic pressure can be equally generated over the circumferential direction of the sliding surface 7a.

In addition, the inner side wall 70a and the outer side wall 70b have a waveshape having a curved surface shape, and thus the movement of the fluid in the dynamic pressure generation groove 70 becomes smooth and dynamic pressure can be stably generated.

In addition, the back pressure chamber 50 extends to the inner diameter side of the sliding surfaces 7a and 8a.

Accordingly, when the sliding surfaces **7a** and **8a** are separated from each other, the fluid in the back pressure chamber **50** is introduced from the inner diameter side of the sliding surfaces **7a** and **8a**. In addition, when the scroll compression mechanism **4** is driven, the pressure of the back pressure chamber **50** increases, a high-pressure fluid is introduced between the sliding surfaces **7a** and **8a** from the back pressure chamber **50**, and thus the sliding surfaces **7a** and **8a** can be further separated from each other by the pressure of the fluid.

In addition, although the present embodiment exemplifies a form in which the radial width dimension L5 (that is, the separation width between the inner side wall **70a** and the outer side wall **70b**) of the dynamic pressure generation groove **70** is constant over the entire circumference, the present invention is not limited thereto. In an alternative disposition, the radial width dimension of the dynamic pressure generation groove may be different at different positions in the circumferential direction of the dynamic pressure generation groove. For example, the inner side wall and the outer side wall may be formed in waveshapes with different amplitudes.

Second Embodiment

Next, a dynamic pressure generation groove **170** of a side seal **107** as a sliding component according to a second embodiment of the present invention will be described with reference to FIG. 6. It should be noted that the description of configurations identical to those of the first embodiment will be omitted for redundancy avoidance.

As for the side seal **107** in the second embodiment illustrated in FIG. 6, a plurality of the dynamic pressure generation grooves **170** extending in an arc shape in the circumferential direction are disposed apart from each other in the circumferential direction of a sliding surface **107a**.

According to this, dynamic pressure can be generated in both circumferential end portions **170d** and **170e** of each dynamic pressure generation groove **170** in addition to an inner side wall **170a** and an outer side wall **170b** of each dynamic pressure generation groove **170**. It should be noted that in this case, it is preferable that the dynamic pressure generation grooves **170** are equally arranged in the circumferential direction.

Third Embodiment

Next, a dynamic pressure generation groove **270** of a side seal **207** as a sliding component according to a third embodiment of the present invention will be described with reference to FIG. 7. It should be noted that the description of configurations identical to those of the first embodiment will be omitted for redundancy avoidance.

As for the dynamic pressure generation groove **270** of the side seal **207** in the third embodiment illustrated in FIG. 7, an inner side wall **270a** is wavy when viewed from the axial direction. In addition, an outer side wall **270b** is circular when viewed from the axial direction. In this manner, the dynamic pressure generation groove may be wavy on at least one of the inner and outer side walls.

In addition, the inner side wall **270a** has an outer diameter side apex **275** that tapers with an acute angle. In addition, the inner side wall **270a** has a waveshape. According to this, the circumferential width of the outer diameter side apex **275** can be reduced, and thus it is possible to ensure a large intersection surface **271** that has a circular arc shape, is

convex to the inner diameter side, and is formed between the outer diameter side apexes **275** adjacent to each other.

Although embodiments of the present invention have been described above with reference to the drawings, the specific configurations are not limited to the embodiments and any changes or additions within the scope of the present invention are included in the present invention.

For example, although a form in which the inner and outer side walls have a waveshape having a curved surface shape has been exemplified in the first to third embodiments, the present invention is not limited thereto and the waveshape may be, for example, rectangular or triangular when viewed from the axial direction.

In addition, although an aspect in which the side seal **7** as a sliding component is applied to the scroll compressor **C** used in the air conditioning system of an automobile or the like has been described in the first to third embodiments, the present invention is not limited thereto and the side seal **7** as a sliding component may be applied to, for example, a scroll expansion compressor provided integrally with an expander and a compressor insofar as it is a rotating machine including an eccentric mechanism.

In addition, each of the fluids in the spaces inside and outside the sliding surface of the sliding component may be any of a gas, a liquid, and a gas-liquid mixture.

In addition, insofar as the sliding component of the present invention has a sliding surface that relatively slides with eccentric rotation, the sliding component of the present invention may be used in an environment in which the pressure inside the sliding surface and the pressure outside the sliding surface are equal to each other without being limited to an environment in which there is a pressure difference between the inside and outside of the sliding surface. In addition, the sliding component of the present invention does not have to function as a seal and may be one capable of stably reducing the frictional resistance of a sliding surface.

In addition, although the side seal having the relatively sliding surface is made of resin and the thrust plate is made of metal in the first to third embodiments, the material of the sliding component may be freely selected in accordance with the environment of use and so on.

In addition, although an aspect in which a dynamic pressure generation groove is formed in the sliding surface of the side seal has been described in the first to third embodiments, the present invention is not limited thereto. A dynamic pressure generation groove may be formed in the sliding region of the sliding surface of the thrust plate (see FIG. 4), which is a sliding component having a sliding surface relatively sliding with eccentric rotation. In addition, dynamic pressure generation grooves may be formed in both the sliding surface of the side seal and the sliding surface of the thrust plate.

In addition, although a configuration in which the sliding surface of the side seal and the sliding surface of the thrust plate as a sliding component slide relative to each other with eccentric rotation has been described in the first to third embodiments, the present invention is not limited thereto. A dynamic pressure generation groove may be formed in the sliding surface relatively sliding with eccentric rotation with only one of the side seal and the thrust plate provided. For example, in a case where only the thrust plate is provided, dynamic pressure generation grooves may be formed in one or both of the sliding surface of the thrust plate as a sliding component and the back surface of the end plate of the movable scroll. In addition, in a case where only the side seal is provided, a dynamic pressure generation groove may

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be formed in the sliding surface of the side seal as a sliding component. In this case, the side seal also functions as a thrust bearing that abuts against the inner peripheral surface of the inner casing and receives the axial load of the movable scroll.

In addition, in a case where the side seal and the thrust plate are not provided and the back surface of the end plate of the movable scroll functions as a thrust bearing that abuts against the inner peripheral surface of the inner casing and receives the axial load of the movable scroll, a dynamic pressure generation groove may be formed in the sliding surface formed on the back surface of the end plate of the movable scroll.

In addition, the sliding surface may be provided with a conduction groove for conduction between the external space on the inner diameter side or the outer diameter side of the sliding surface and the dynamic pressure generation groove.

In addition, although a form in which the external space on the low-pressure side is on the outer diameter side of the side seal and the high-pressure external space is on the inner diameter side of the side seal has been exemplified, the external space on the low-pressure side may be on the inner diameter side of the side seal with the high-pressure external space on the outer diameter side of the side seal.

REFERENCE SIGNS LIST

- 4 Scroll compression mechanism
- 7 Side seal (sliding component)
- 7a Sliding surface
- 8 Thrust plate
- 8a Sliding surface
- 20 Low-pressure chamber (external space on outer diameter side)
- 30 High-pressure chamber
- 40 Compression chamber
- 41 Fixed scroll
- 42 Movable scroll
- 50 Back pressure chamber (external space on inner diameter side, external space on high-pressure side)
- 70 Dynamic pressure generation groove
- 70a Inner side wall (side wall on inner diameter side)
- 70b Outer side wall (side wall on outer diameter side)
- 107 Side seal (sliding component)
- 107a Sliding surface
- 170 Dynamic pressure generation groove
- 170a Inner side wall (side wall on inner diameter side)
- 170b Outer side wall (side wall on outer diameter side)
- 207 Side seal (sliding component)
- 270a Inner side wall (side wall on inner diameter side)
- 270b Outer side wall (side wall on outer diameter side)
- 271 Intersection surface
- 701 to 704 Intersection surface
- C Scroll compressor
- M Drive motor

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The invention claimed is:

1. An eccentric sliding assembly comprising: a first sliding component having a sliding surface; a second sliding component having a sliding surface slidable to the sliding surface of the first sliding component; a rotational member configured to rotate around a predetermined axis; a movable member which is the first sliding component or to which the first sliding component is attached;

and a fixed member which is the second sliding component or to which the second sliding component is attached; wherein the rotational member has an eccentric portion eccentrically formed with respect to the predetermined axis and protruded toward the movable member, the movable member has a boss fitted around the eccentric portion of the rotational member and is relatively rotatable therewith, and the sliding surface of at least one of the first sliding component and the second sliding component is provided with a dynamic pressure generation groove defined by side walls extending in a circumferential direction, at least one of the side walls being formed in a waveshape with amplitude in a radial direction.

2. The eccentric sliding assembly according to claim 1, wherein both the side walls of the dynamic pressure generation groove are formed in the waveshape with the amplitude in the radial direction.

3. The eccentric sliding assembly according to claim 1, wherein the dynamic pressure generation groove is formed in an annular shape.

4. The eccentric sliding assembly according to claim 1, wherein the waveshape of the side wall is periodically formed with a constant pitch.

5. The eccentric sliding assembly according to claim 1, wherein the waveshape of the side wall is formed by curved surfaces.

6. The eccentric sliding assembly according to claim 2, wherein the dynamic pressure generation groove is formed in an annular shape.

7. The eccentric sliding assembly according to claim 2, wherein the waveshape of the side wall is periodically formed with a constant pitch.

8. The eccentric sliding assembly according to claim 2, wherein the waveshape of the side wall is formed by curved surfaces.

9. The eccentric sliding assembly according to claim 3, wherein the waveshape of the side wall is periodically formed with a constant pitch.

10. The eccentric sliding assembly according to claim 3, wherein the waveshape of the side wall is formed by curved surfaces.

11. The eccentric sliding assembly according to claim 4, wherein the waveshape of the side wall is formed by curved surfaces.

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